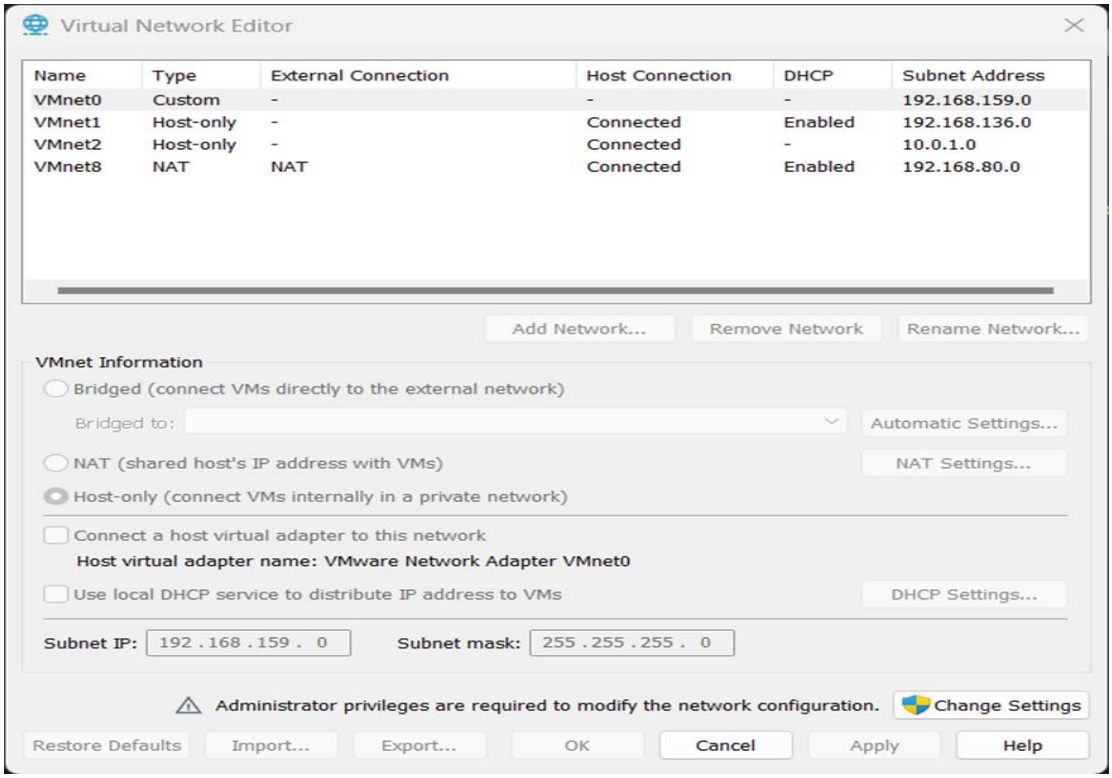
Lab Sheet 2

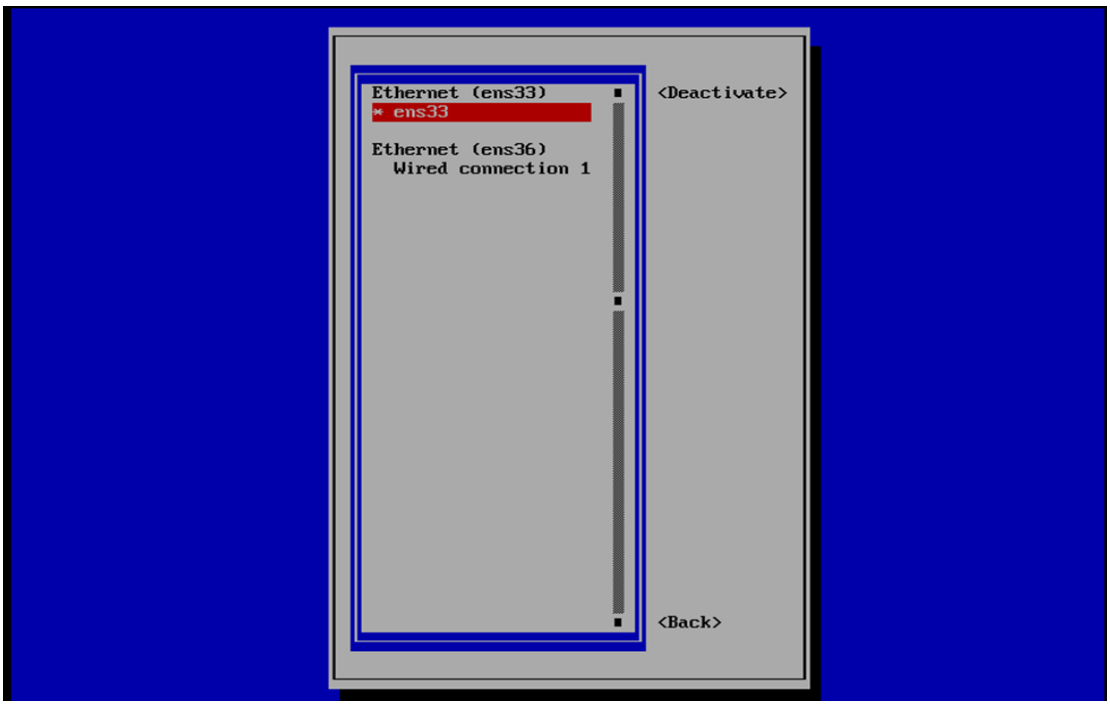
IT Number: IT22548078

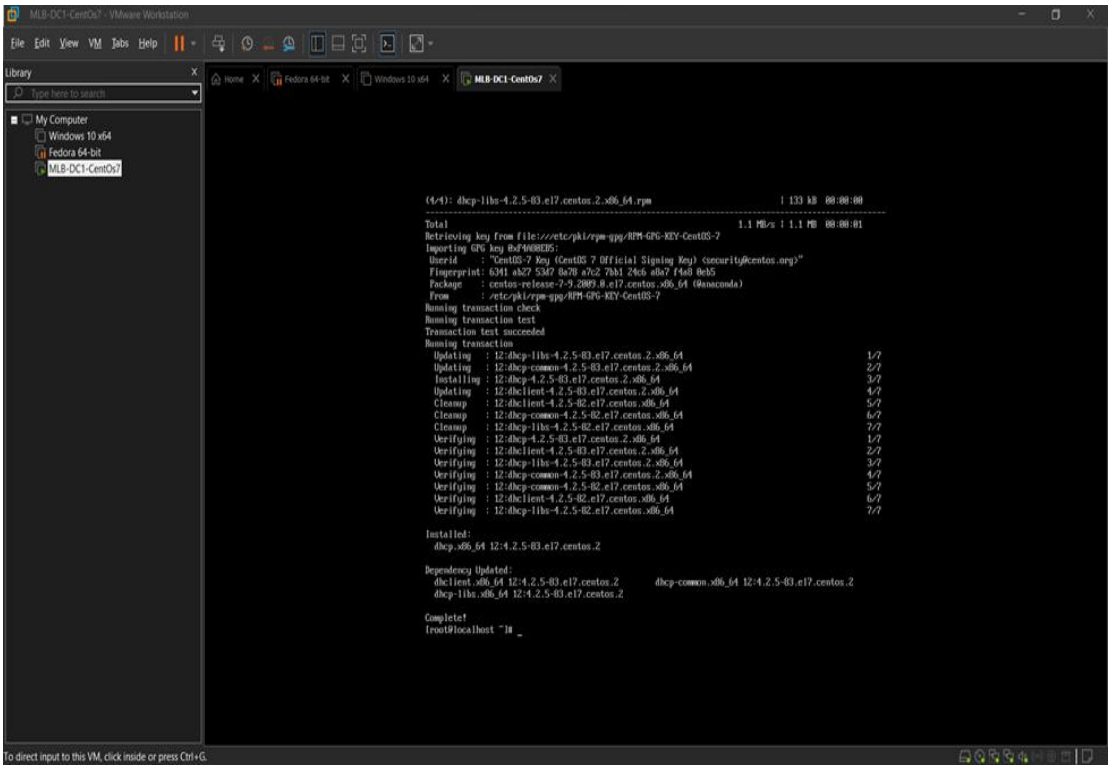
Name: M.T.K.Chathuri

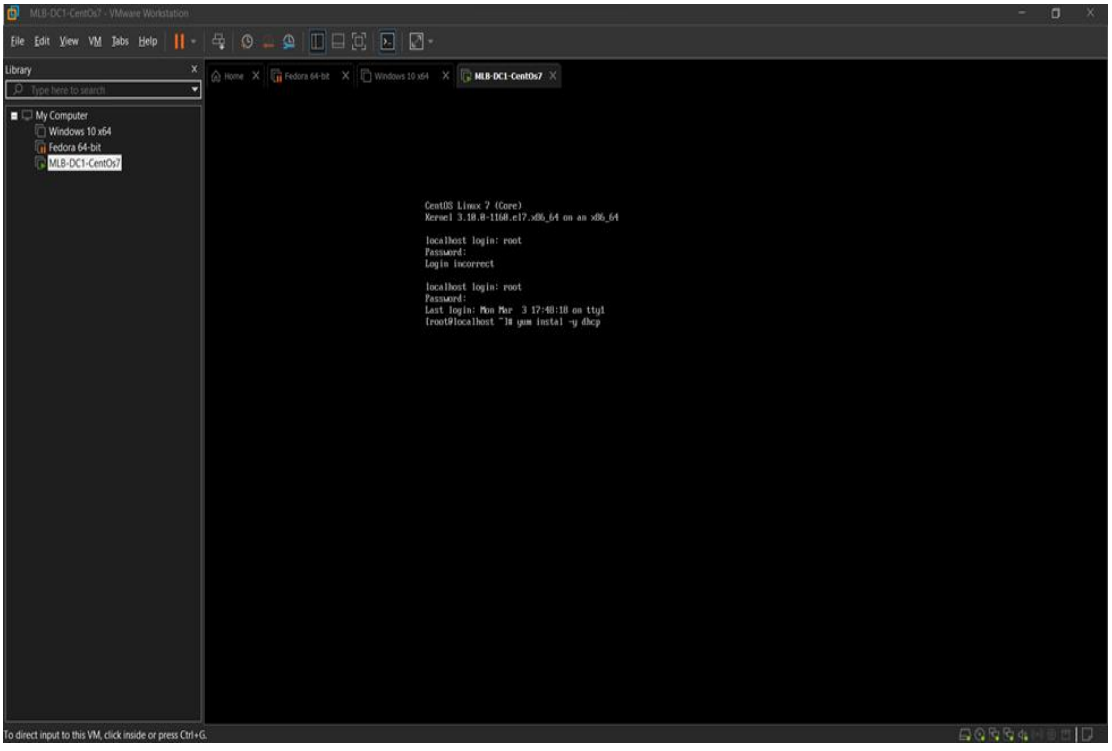
Batch: WE.IT.0202

Screenshot Analysis






MLB-DC1-CentOs7 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Windows 10 x64
Fedora 64-bit
MLB-DC1-CentOs7
Home
Fedora 64-bit
Windows 10 x64
MLB-DC1-CentOs7
(4/4): dhcp-libs-4.2.5-83.el7.centos.2.x86_64.rpm | 133 kB 00:00:00
Total 1.1 MB/s | 1.1 MB 00:00:01
Retrieving key from file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
Importing GPG key 0xF4A80EB5:
Userid : "CentOS-7 Key (CentOS 7 Official Signing Key) <security@centos.org>"
Fingerprint: 6341 ab27 53d7 8a78 a7c2 7bb1 24c6 a8a7 f4a8 0eb5
Package : centos-release-7-9.2009.0.el7.centos.x86_64 (@anaconda)
From : /etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Updating : 12:dhcp-libs-4.2.5-83.el7.centos.2.x86_64 1/7
Updating : 12:dhcp-common-4.2.5-83.el7.centos.2.x86_64 2/7
Installing : 12:dhcp-4.2.5-83.el7.centos.2.x86_64 3/7
Updating : 12:dhclient-4.2.5-83.el7.centos.2.x86_64 4/7
Cleanup : 12:dhclient-4.2.5-82.el7.centos.x86_64 5/7
Cleanup : 12:dhcp-common-4.2.5-82.el7.centos.x86_64 6/7
Cleanup : 12:dhcp-libs-4.2.5-82.el7.centos.x86_64 7/7
Verifying : 12:dhcp-4.2.5-83.el7.centos.2.x86_64 1/7
Verifying : 12:dhclient-4.2.5-83.el7.centos.2.x86_64 2/7
Verifying : 12:dhcp-libs-4.2.5-83.el7.centos.2.x86_64 3/7
Verifying : 12:dhcp-common-4.2.5-83.el7.centos.2.x86_64 4/7
Verifying : 12:dhcp-common-4.2.5-82.el7.centos.x86_64 5/7
Verifying : 12:dhclient-4.2.5-82.el7.centos.x86_64 6/7
Verifying : 12:dhcp-libs-4.2.5-82.el7.centos.x86_64 7/7
Installed:
dhcp.x86_64 12:4.2.5-83.el7.centos.2
Dependency Updated:
dhclient.x86_64 12:4.2.5-83.el7.centos.2 dhcp-common.x86_64 12:4.2.5-83.el7.centos.2
dhcp-libs.x86_64 12:4.2.5-83.el7.centos.2
Complete!
[root@localhost ~]# _
To direct input to this VM, click inside or press Ctrl+G.


MLB-DC1-CentOs7 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Windows 10 x64
Fedora 64-bit
MLB-DC1-CentOs7
Home
Fedora 64-bit
Windows 10 x64
MLB-DC1-CentOs7
CentOS Linux 7 (Core)
Kernel 3.10.0-1160.el7.x86_64 on an x86_64
localhost login: root
Password:
Login incorrect
localhost login: root
Password:
Last login: Mon Mar 3 17:48:18 on tty1
[root@localhost ~]# yum instal -y dhcp
To direct input to this VM, click inside or press Ctrl+G.

```
# WARNING: This file is NOT used anymore.

# If you are here to restrict what interfaces should dhcpd listen on,
# be aware that dhcpd listens *only* on interfaces for which it finds subnet
# declaration in dhcpd.conf. It means that explicitly enumerating interfaces
# also on command line should not be required in most cases.

# If you still insist on adding some command line options,
# copy dhcpd.service from /lib/systemd/system to /etc/systemd/system and modify
# it there.
# https://fedoraproject.org/wiki/Systemd#How_do_I_customize_a_unit_file.2F_add_a_custom_unit_file.3F

# example:
# $ cp /usr/lib/systemd/system/dhcpd.service /etc/systemd/system/
# $ vi /etc/systemd/system/dhcpd.service
# $ ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid <your_in
terface_name(s)>
# $ systemctl --system daemon-reload
# $ systemctl restart dhcpd.service
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
"/etc/sysconfig/dhcpd" 18L, 890C
```

```
# If you are here to restrict what interfaces should dhcpd listen on,
# be aware that dhcpd listens *only* on interfaces for which it finds subnet
# declaration in dhcpd.conf. It means that explicitly enumerating interfaces
# also on command line should not be required in most cases.

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# copy dhcpd.service from /lib/systemd/system to /etc/systemd/system and modify
# it there.
# https://fedoraproject.org/wiki/Systemd#How_do_I_customize_a_unit_file.2F_add_a_custom_unit_file.3F

# example:
# $ cp /usr/lib/systemd/system/dhcpd.service /etc/systemd/system/
# $ vi /etc/systemd/system/dhcpd.service
# $ ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid <your_in
terface_name(s)>
# $ systemctl --system daemon-reload
# $ systemctl restart dhcpd.service

DHCPDARGS=ens36
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
"/etc/sysconfig/dhcpd" 20L, 907C written
[root@localhost ~]#
[root@localhost ~]# cp /usr/share/doc/dhcp*/dhcpd.conf.example /etc/dhcp/dhcpd.conf
cp: overwrite ‘/etc/dhcp/dhcpd.conf’? yes
[root@localhost ~]#
```

```
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#

# option definitions common to all supported networks...
option domain-name "example.org";
option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;
max-lease-time 7200;

# Use this to enble / disable dynamic dns updates globally.
#ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
#authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
log-facility local7;

# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

subnet 10.152.187.0 netmask 255.255.255.0 {
}

# This is a very basic subnet declaration.

subnet 10.254.239.0 netmask 255.255.255.224 {
  range 10.254.239.10 10.254.239.20;
  option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
}

"/etc/dhcp/dhcpd.conf" 104L, 3262C
```

```
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#

# option definitions common to all supported networks...
option domain-name "dsnm.sub";
option domain-name-servers server.unixmen.local;

default-lease-time 600;
max-lease-time 7200;

# Use this to enble / disable dynamic dns updates globally.
#ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
log-facility local7;

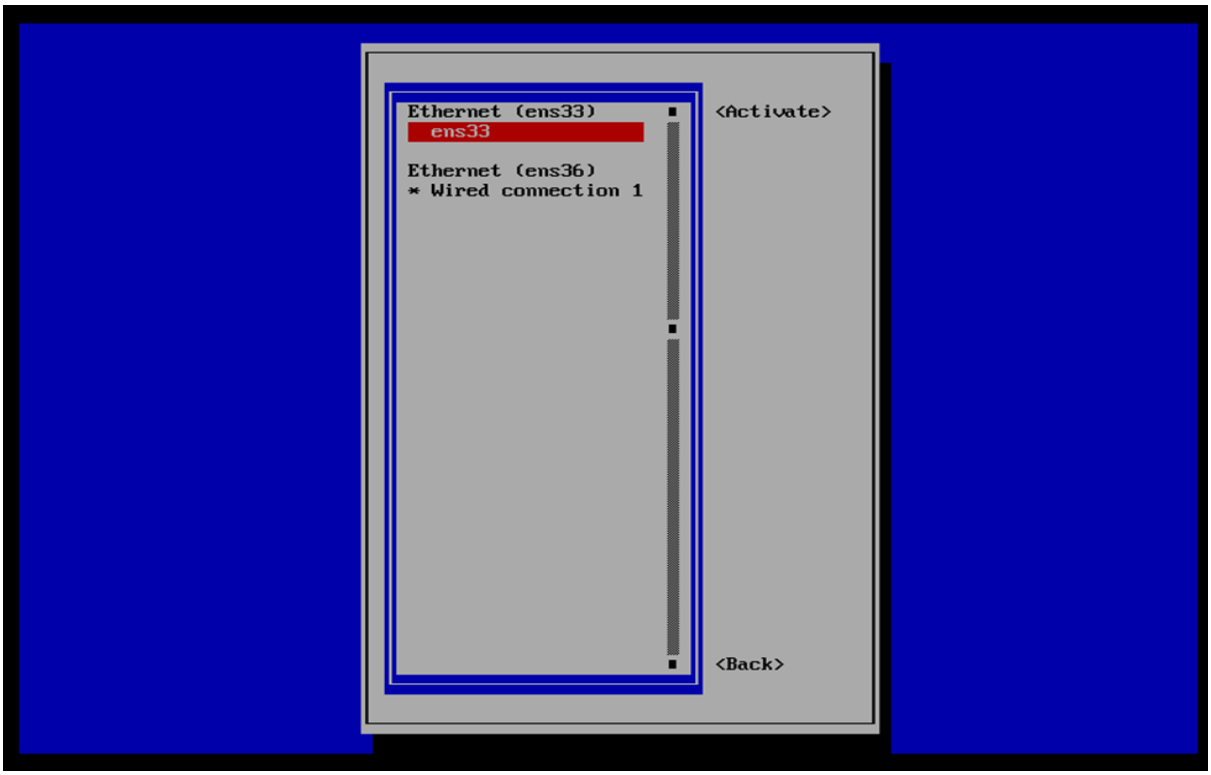
# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

subnet 10.152.187.0 netmask 255.255.255.0 {
}

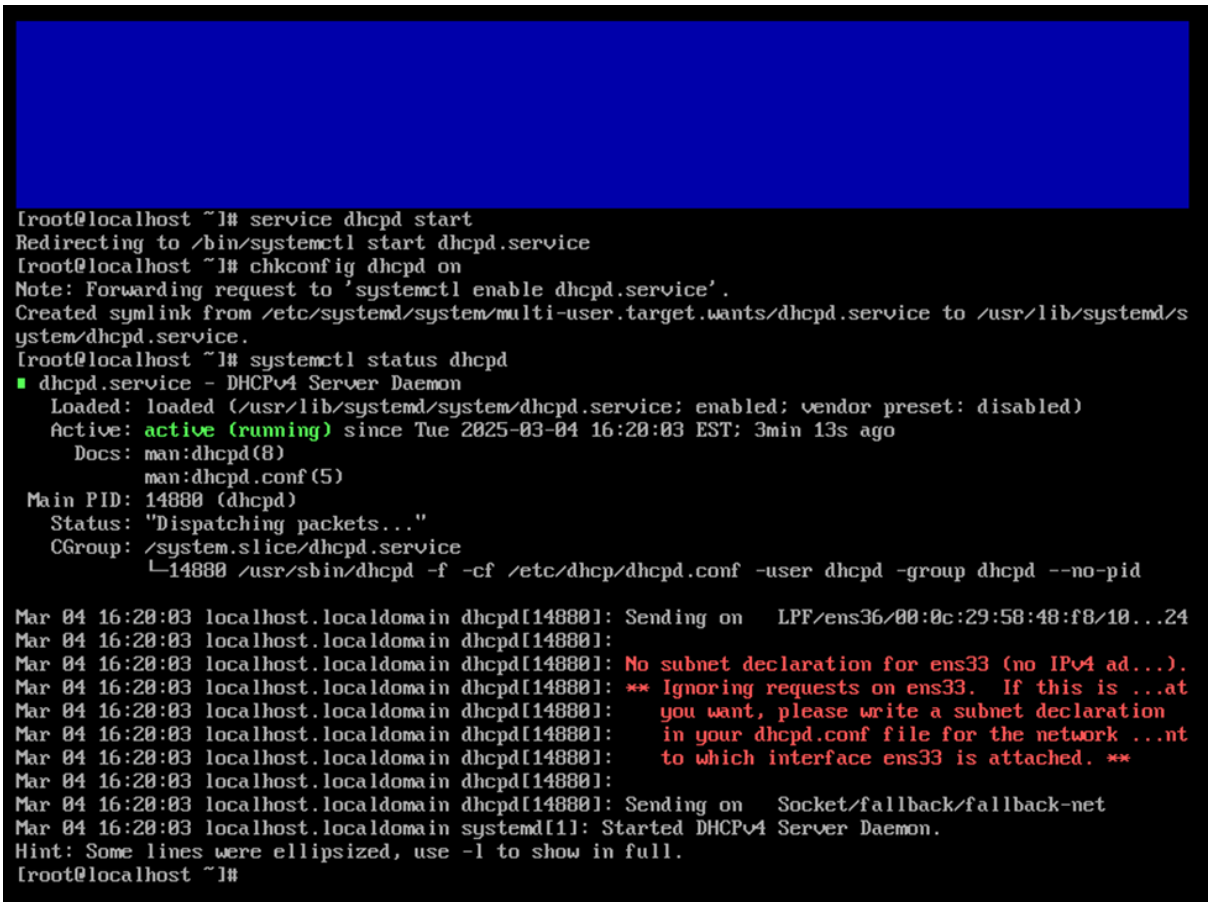
# This is a very basic subnet declaration.

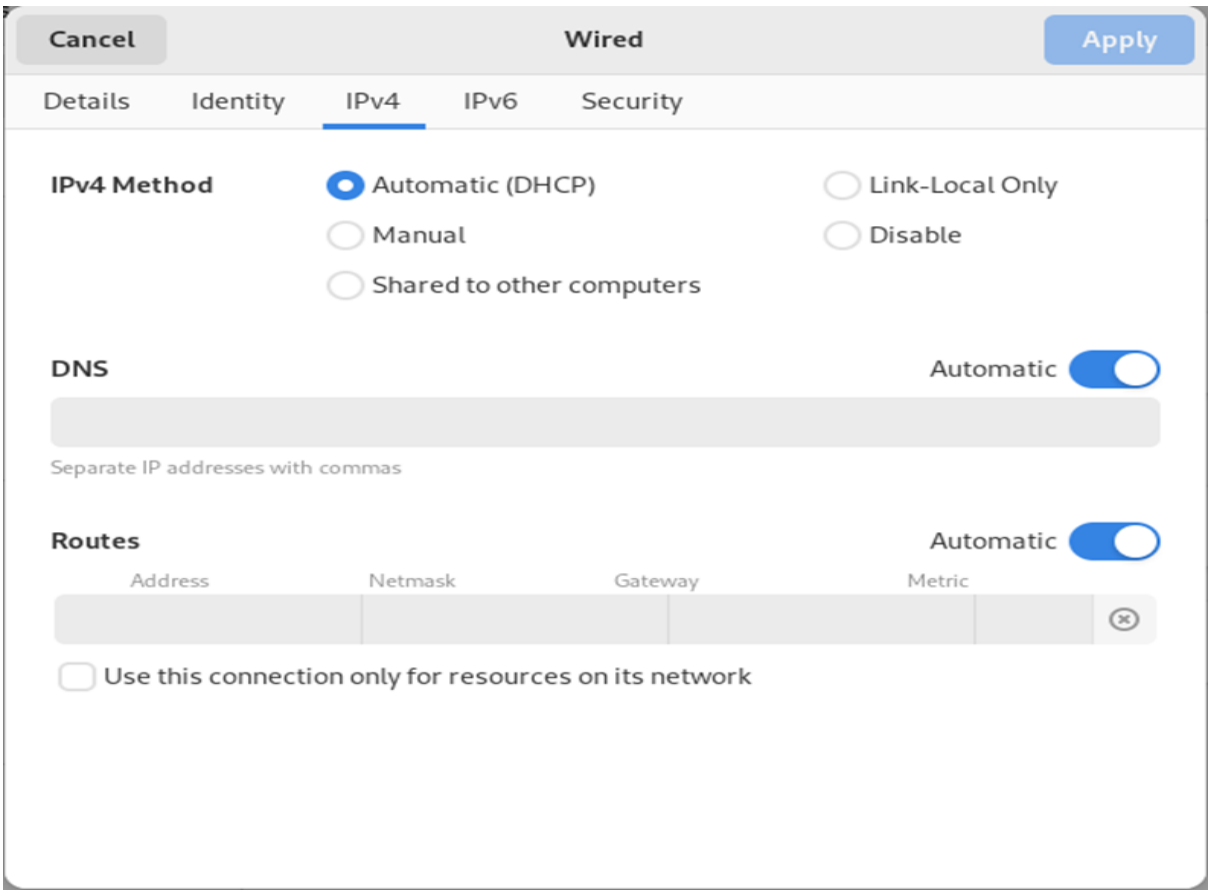
subnet 10.254.239.0 netmask 255.255.255.224 {
  range 10.254.239.10 10.254.239.20;
  option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
}

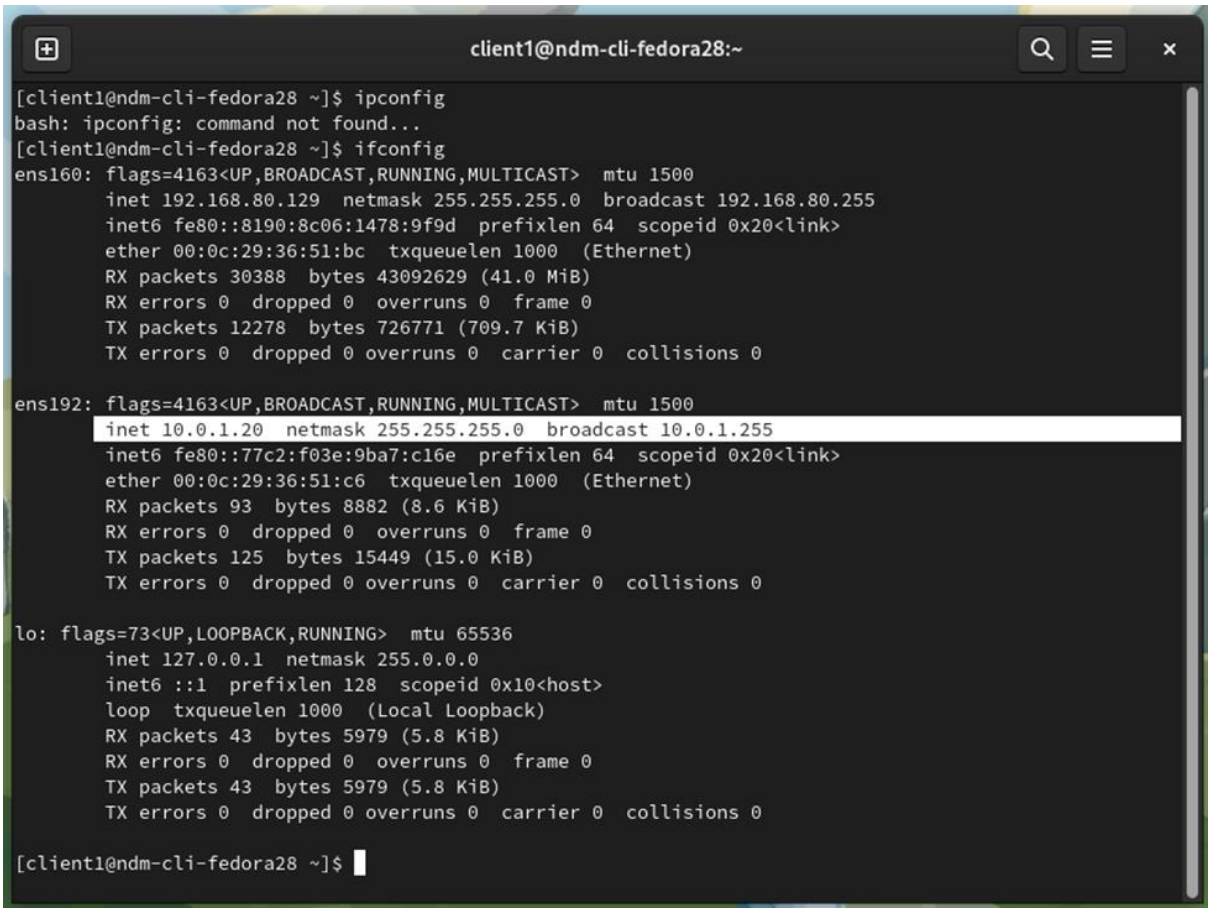
-- INSERT --
```


Ethernet (ens33)
ens33
Ethernet (ens36)
* Wired connection 1
<Activate>
<Back>


[root@localhost ~]# service dhcpd start
Redirecting to /bin/systemctl start dhcpd.service
[root@localhost ~]#


[root@localhost ~]# service dhcpd start
Redirecting to /bin/systemctl start dhcpd.service
[root@localhost ~]# chkconfig dhcpd on
Note: Forwarding request to 'systemctl enable dhcpd.service'.
Created symlink from /etc/systemd/system/multi-user.target.wants/dhcpd.service to /usr/lib/systemd/system/dhcpd.service.
[root@localhost ~]# systemctl status dhcpd
dhcpd.service - DHCPv4 Server Daemon
Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; vendor preset: disabled)
Active: active (running) since Tue 2025-03-04 16:20:03 EST; 3min 13s ago
Docs: man:dhcpd(8)
man:dhcpd.conf(5)
Main PID: 14880 (dhcpd)
Status: "Dispatching packets..."
CGroup: /system.slice/dhcpd.service
└─14880 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]: Sending on LPF/ens36/00:0c:29:58:48:f8/10...24
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]:
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]: No subnet declaration for ens33 (no IPv4 ad...).
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]: ** Ignoring requests on ens33. If this is ...at
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]: you want, please write a subnet declaration
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]: in your dhcpd.conf file for the network ...nt
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]: to which interface ens33 is attached. **
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]:
Mar 04 16:20:03 localhost.localdomain dhcpd[14880]: Sending on Socket/fallback/fallback-net
Mar 04 16:20:03 localhost.localdomain systemd[1]: Started DHCPv4 Server Daemon.
Hint: Some lines were ellipsized, use -l to show in full.
[root@localhost ~]#


Cancel
Wired
Apply
Details
Identity
IPv4
IPv6
Security
IPv4 Method
Automatic (DHCP)
Link-Local Only
Manual
Disable
Shared to other computers
DNS
Automatic
Separate IP addresses with commas
Routes
Automatic
Address
Netmask
Gateway
Metric
Use this connection only for resources on its network




```
# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

subnet 10.152.187.0 netmask 255.255.255.0 {
}

# This is a very basic subnet declaration.

subnet 10.254.239.0 netmask 255.255.255.224 {
  range 10.254.239.10 10.254.239.20;
  option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
}

# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.

subnet 10.254.239.32 netmask 255.255.255.224 {
  range dynamic-bootp 10.254.239.40 10.254.239.60;
  option broadcast-address 10.254.239.31;
  option routers rtr-239-32-1.example.org;
}

# A slightly different configuration for an internal subnet.
subnet 10.0.1.0 netmask 255.255.255.0 {
  range 10.0.1.20 10.0.1.30;
  option domain-name-servers server.unixmen.local;
  option domain-name "dsnm.sub";
  option routers 10.0.1.1;
  option broadcast-address 10.0.1.255;
  default-lease-time 600;
  max-lease-time 7200;
}

# Hosts which require special configuration options can be listed in
# host statements.   If no address is specified, the address will be
"/etc/dhcp/dhcpd.conf" 104L, 3229C written
[root@localhost ~]#
```